

RESEARCH ARTICLE

Synergies Within and Potential Barriers to Global Forest Sustainable Development

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ABSTRACT

Amid the escalating global environmental crisis and the pressing imperative to advance the sustainable development goals (SDGs), forest sustainability has emerged as a critical nexus between ecological conservation and economic development. The key drivers and constraints that shape global forest sustainability are not yet fully understood. We developed a forest sustainability index (FSI) that integrates both ecological and socio-economic dimensions to assess spatial and temporal trends across 73 countries. The results indicate a modest global improvement in forest sustainability, with the average FSI increasing from 0.187 to 0.191 in 2010–2020. However, South America—particularly Argentina and Chile—exhibited persistently high FSI, forming sustainability hotspots. In contrast, Northern Africa and parts of Eastern and Northern Europe consistently emerged as coldspots, characterized by chronically low FSI and heightened vulnerability. The coupling coordination degree (CCD) between ecological and socio-economic factors was significantly stronger in hotspots (CCD = 0.31) than in coldspots (CCD = 0.24), where the CCD remained in a state of severe dysfunction. Barrier analysis identified limited forest stocking capacity as the primary constraint on system synergy, while precipitation and forest-related carbon emissions emerged as the dominant natural drivers influencing sustainable development. These findings underscore the pivotal role of natural environmental factors in enabling or impeding progress toward forest sustainability. The study provides critical insights for designing region-specific forest governance strategies and provides a scientific basis for accelerating progress toward SDG 15 (life on land).

1 | Introduction

The United Nations (UN) 2030 Agenda for Sustainable Development identifies “Sustainable Forest Management” as a core objective, underscoring the essential role of forests in mitigating climate change and promoting socio-economic equity (UN 2015; Hu et al. 2025). Assessing forest sustainability across countries not only revealed overarching trends but also highlights interregional disparities—insights that are vital for informing national policy adjustments (Keenan

et al. 2015). The international community increasingly recognizes the urgent need for sustainable forest governance, particularly in reconciling the oftencompeting priorities of ecological preservation and socio-economic development, as highlighted in recent discussions on forest–energy trade-offs (Hansmann 2024). Given the inherent complexity and dynamism of forest ecosystems, shaped by interactions among diverse biotic and abiotic factors, comprehensive assessments must integrate ecological, economic, and social dimensions (Frini and Ben Amor 2019). Heterogeneity in forest resource

and institutional contexts remains a major challenge to developing standardized assessment methodologies (Wang et al. 2023). Consequently, quantifying forest sustainability in a globally consistent and meaningful way remains a significant scientific and policy challenge.

A standardized indicator system is essential for comparing forest management practices across countries and promoting international cooperation. Evaluating forest sustainability globally aligns with current research priorities. While many studies focus on developing indicators and regional assessment frameworks (Mattsson et al. 2024), few apply composite indices at the national level (Baral et al. 2018; Pietarinen et al. 2023; Hasnaoui and Krott 2019). The FAO's global forest resources assessment (FRA) remains the most comprehensive, integrating forest quality and benefits for global monitoring (FAO 2020). However, most assessments remain national in scope, limiting cross-country comparability. To address this, there is a need for an integrated global framework using remote sensing, monitoring systems, and geographic data (Sarkissian and Kutia 2024).

Sustainability is defined as a dynamic process or condition that can be maintained over time, encompassing three interrelated dimensions: ecological, economic, and social (Martínez-Vega et al. 2016). Forest sustainability specifically involves not only the enhanced development of socio-economic systems but also the preservation of healthy and resilient forest ecosystems (Vatandaslar et al. 2024). While a substantial body of research has explored frameworks and driving forces related to forest sustainability (Roshani et al. 2022; Wu and Liu 2024), critical gaps remain in understanding how these drivers interact across contexts. Among natural drivers, climate change is widely recognized as a major factor influencing forest structure and ecosystem functionality (Forzieri et al. 2022; Liu et al. 2023). Human activities, such as reforestation and unsustainable resource extraction, also represent key anthropogenic pressures (Grantham et al. 2020). Conversely, sustainable management measures that integrate conservation and restoration, such as forest certification schemes and community-based forest management, offer promising solutions for advancing forest sustainability (Kant and Berry 2005; Bhatt 2022; Borges-Matos et al. 2025). Moreover, robust policy frameworks and effective enforcement mechanisms have been shown to significantly improve forest resource protection and support sustainable forest management (Pukkala 2021). It is critical to identify the constraints to forest sustainability and identify coordination barriers.

The rapid advancement of remote sensing technologies has significantly enhanced large-scale forest monitoring by offering cost-effective, efficient, and comprehensive data acquisition (Pettorelli et al. 2005). Remote sensing data now serve as a critical foundation for evaluating forest resources and monitoring forest dynamics on a global scale (Hansen et al. 2013). With the integration of geospatial analysis and vegetation indicators, researchers can more accurately track forest cover changes and assess forest sustainability across spatial and temporal scales (Khunrattanasiri 2022; D'Agata et al. 2025; Takahashi and Todo 2014). Building on these technological advancements, this study constructs a FSI by integrating FAO statistics with

multi-source geospatial and thematic datasets. Previous global monitoring systems, such as those under the FRA, have focused mainly on ecological indicators like forest cover or biomass, often overlooking socio-economic and institutional dimensions. In contrast, the FSI developed here includes both ecological and socio-economic aspects, supported by geographic data, statistical information, and monitoring technologies. This integrated approach ensures broad spatial coverage and comparability across countries, offering deeper insights for policymaking and international benchmarking. It enables cross-country comparisons that reflect both environmental conditions and the social systems that govern forests, offering valuable insights for forestry policy development and international benchmarking.

Specifically, this study addresses the following questions: (1) How sustainable are forests in major forested countries around the world? (2) Where are the coldspot and hotspot patterns of forest sustainability globally? (3) How are the ecological and socio-economic subsystems of forest sustainability coordinated in coldspot and hotspot regions, and what factors constrain their coordination? (4) What are the main natural and human-related drivers influencing forest sustainability?

2 | Data and Methods

2.1 | Data Sources

The UN's FAO has published comprehensive Global FRA reports every 5 years since 1980, offering authoritative, longitudinal data on global forest resources spanning the period from 1990 to 2020 (FAO 2020). These reports provide critical input for international frameworks such as the convention on biological diversity (CBD) and the United Nations Forum on Forests (UNFF), serving as a foundational reference for forest policy and sustainability assessments worldwide.

This study primarily utilized panel data from the FAO's FRA reports, given their credibility and extensive coverage. While a coordinated indicator framework holds strong theoretical advantages for assessing forest sustainability, its practical implementation was often constrained by challenges such as data availability, indicator harmonization, and temporal consistency. To address these limitations and enhance analytical robustness, we selected 73 countries with relatively complete and high-quality datasets to conduct quantitative evaluations of forest sustainability (Table A1). This selection ensures the representativeness and scientific rigor of the study findings. In addition to statistical panel data, this study incorporates satellite-based remote sensing indicators, which are less influenced by differences in national statistical standards and methodologies, thereby reducing data heterogeneity. Specifically, raster datasets such as the normalized difference vegetation index (NDVI), net primary productivity (NPP), and leaf area index (LAI) were used as core indicators for ecological performance (Alcaraz-Segura et al. 2009; Lu et al. 2023; Guo et al. 2023; Dong et al. 2024). These indicators are widely recognized for their ability to capture vegetation growth dynamics, ecosystem productivity, and canopy structural attributes, making them essential metrics for assessing the ecological dimension of forest sustainability and sustainable forest management practices (Figure 1).

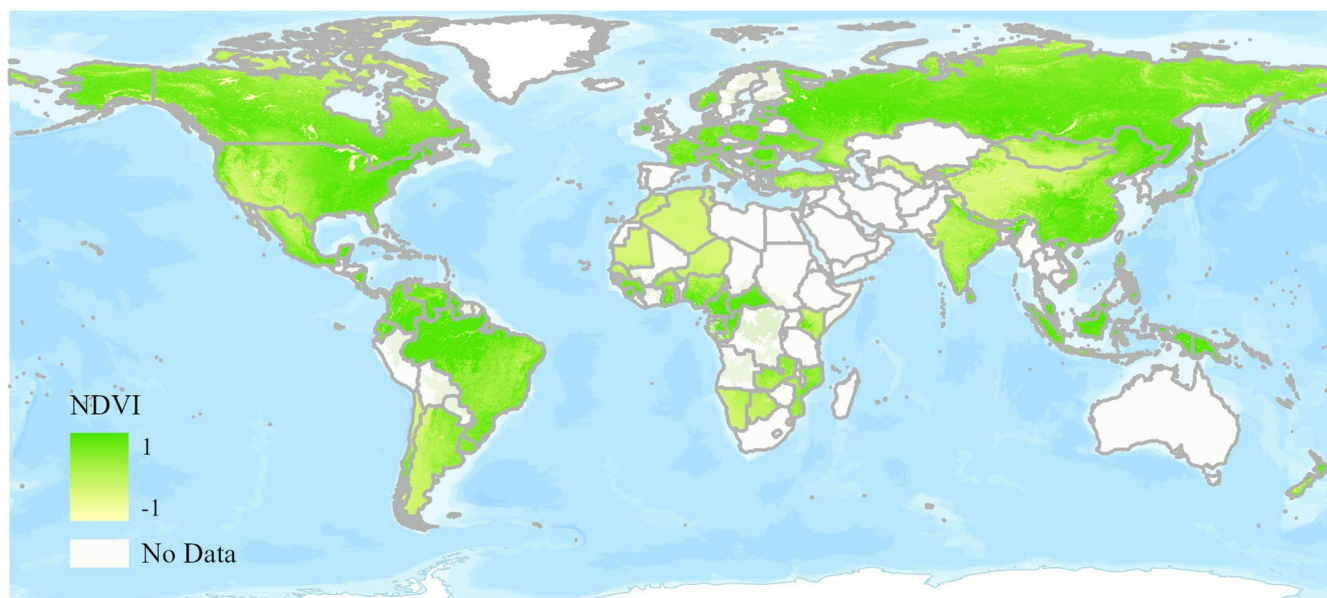


FIGURE 1 | Geographic distribution and NDVI representation of the 73 study countries. (This map illustrates the spatial extent of the 73 countries included in the analysis, overlaid with normalized difference vegetation index (NDVI) data to represent vegetation greenness and density. Areas shown in white or left blank indicate regions not included in the study).

2.2 | Forest Sustainability Evaluation Indicators

The selection of indicators in this study followed three core principles: (1) each indicator exhibited a clear monotonic relationship with at least one aspect of forest sustainability; (2) the indicator possessed sufficient temporal and spatial data coverage across multiple countries and years; and (3) the indicator was simple, transparent, and easily processable (Zhang et al. 2021). Adhering to these principles, and ensuring data were available at the required spatial and temporal resolutions for practical cross-national comparisons, we consulted a range of authoritative sources, including peer-reviewed literature (Martínez-Vega et al. 2016), the European System of Sustainability Indicators (Linser and Wolfslehner 2022; Rametsteiner et al. 2011), and the FAO's forest sustainability framework (FAO 2020). These resources guided the selection and structuring of forest sustainability indicators.

Regarding data preprocessing, statistical indicators were primarily derived from the FAO's global forest resources assessment, and normalization was applied to ensure comparability across variables. For remote sensing-based metrics—namely the NDVI, LAI, and NPP—we calculated the average values for the years 2010, 2015, and 2020. These values were then aggregated to produce mean national level estimates for each of the 73 countries during the study period. Based on the FAO's Forest Sustainability Framework and in alignment with integrated sustainability theory, 17 indicators were selected to construct the FSI system. To emphasize the holistic nature of forest sustainability, particularly the interplay between ecological and anthropogenic pressures, 17 indicators were classified into two overarching dimensions: the natural-ecological dimension and the socio-economic dimension (Table 1).

2.3 | Drivers of Forest Sustainability

Climate change and human activities can affect the development of forest sustainability. We identified and constructed the drivers of FSI based on the underlying mechanisms of FSI change, a review of relevant literature, and the natural ecological and socio-economic context of FSI (Payn et al. 2015; Forzieri et al. 2022; Chang et al. 2025). The driver evaluation system was divided into two main categories: natural and anthropogenic. Drivers were selected from both dimensions according to the principles of causal relevance, representativeness, and quantifiability (Kpadé et al. 2024). A total of eight factors were ultimately identified as the main drivers of forest sustainability (Table 2).

2.4 | Research Methods

With the advancement of sustainable forest management, single indicators are no longer sufficient to capture the complexity of forest sustainability. We developed an integrated assessment framework that combines ecological and socio-economic system indicators to analyze the spatial distribution, coordination patterns, and driving mechanisms of global forest sustainability (Figure 2).

2.4.1 | Forest Sustainability Assessment Method

As single indicators are insufficient to reflect the complexity of forest sustainability, researchers have increasingly adopted composite evaluation methods (Ding et al. 2016; Xu et al. 2021). In this study, the entropy weight method was applied due to its ability to assign indicator weights objectively based on data variability (Wu et al. 2022). A total of 17 indicators were standardized and weighted to construct the FSI across 73 countries.

TABLE 1 | Indicators of forest sustainability in 73 countries, including classification, definitions, weights, and data sources.

Indicator category (number)	Indicator name	Average indicator weights	Source of data
Natural Ecology (12)	x1: forest area (% of land area)	0.025	https://fra-data.fao.org/
	x2: forest area in protected areas (% of forest area)	0.043	https://fra-data.fao.org/
	x3: forest stock	0.210	https://fra-data.fao.org/
	x4: forest area targeted for biodiversity conservation (% of total forest area)	0.063	https://fra-data.fao.org/
	x5: natural regeneration forest stock (% of total forest stock)	0.009	https://fra-data.fao.org/
	x6: normalized difference vegetation index (NDVI)	0.017	https://www.nasa.gov/
	x7: leaf area index (LAI)	0.020	https://lpdaac.usgs.gov/
	x8: net primary productivity (NPP)	0.035	https://lpdaac.usgs.gov/
	x9: above-ground biomass data	0.035	https://fra-data.fao.org/
	x10: above-ground biomass carbon stock data	0.034	https://fra-data.fao.org/
	x11: other forested land area as % of national territory	0.129	https://fra-data.fao.org/
	x12: forest area change in 5 years (% of existing forest area)	0.003	https://fra-data.fao.org/
Social Economic (5)	x13: total stock of introduced species	0.262	https://fra-data.fao.org/
	x14: platforms to promote or allow stakeholder participation in forest policy development	0.013	https://fra-data.fao.org/
	x15: forest area with long-term conservation plan (% of forest area)	0.038	https://fra-data.fao.org/
	x16: forest area targeted for production (% of forest area)	0.059	https://fra-data.fao.org/
	x17: data on log consumption	0.005	https://www.fao.org/

Note: All percentage-based indicators have been rewritten using the format “%” for clarity. “Other forested land” refers to areas not classified as forest but still meeting specific vegetation thresholds—more than 0.5 ha in size, with trees taller than 5 m and a canopy cover of 5%–10%, or with a combined cover of shrubs, bushes, and trees above 10%. This does not include land primarily under agricultural or urban use, in accordance with FAO FRA 2020 definitions.

Sensitivity analysis further confirmed the robustness of the results (Figure A1).

Positive indicators:

$$X'_{ij} = \frac{X_{ij} - \min X_{ij}}{\max X_{ij} - \min X_{ij}} \quad (1)$$

Negative indicators:

$$X'_{ij} = \frac{\max X_{ij} - X_{ij}}{\max X_{ij} - \min X_{ij}} \quad (2)$$

where X_{ij} was the raw data of the j th indicator for the i th country; X'_{ij} was the value of the corresponding standardised variable, $X'_{ij} \in [0, 1]$; $\max(X_{ij})$, $\min(X_{ij})$ are the maximum and minimum values of the j th indicator, respectively.

2.4.2 | Coupling Coordination Degree Assessment Method of Subsystems

The coupled coordination degree model (CCDM) was used to assess the interaction between the natural ecological and

socio-economic subsystems within the FSI framework. CCDM captures both the degree of coupling and coordination between systems, offering a comprehensive measure of their integrated development (Liu and Zhao 2024; Zhang et al. 2023; Zhu et al. 2026).

$$\text{CCD} = \sqrt{\text{CD} * T} \quad (3)$$

$$\text{CD} = n * \left[\frac{m_1 * m_2 * \dots * m_n}{(m_1 + m_2 + \dots + m_n)^n} \right]^{\frac{1}{n}} \quad (4)$$

$$T = \frac{1}{n} (m_1 + m_2 + \dots + m_n) \quad (5)$$

where CCD ($0 \leq \text{CCD} \leq 1$) denoted the coupling coordination degree, with higher values indicating a greater level of coordinated development within or between systems. CD represents the degree of coupling (D). T was the comprehensive coordination index, m denoted the number of variables within a subsystem, and n was the total number of variables considered. In this study, the natural ecological and socio-economic subsystems of forest sustainability were assumed to be of equal importance.

TABLE 2 | Key drivers of forest sustainability and data sources.

Type of driver (number)	Name of driver	Source of data
Natural factor (5)	A1: digital elevation model (DEM)	https://www.copernicus.eu/en
	A2: emissions from forests	https://www.fao.org/
	A3: temperature change on land	https://www.fao.org/
	A4: loss of forest area due to fire	https://www.fao.org/
	A5: average annual precipitation	https://developers.google.com/
Human factor (3)	A6: GDP per capita	https://www.worldbank.org/
	A7: population density	https://www.worldbank.org/
	A8: occurrence of armed struggle in the country	https://ourworldindata.org/

Therefore, equal weights were assigned to both. CCD was categorized into 10 grades, from 1 (extremely unbalanced, CCD < 0.1) to 10 (superiorly balanced, CCD > 0.9), reflecting the degree of coordination between ecological and socio-economic subsystems (Table A2).

2.4.3 | Method for Identifying Obstacle Factors to Coupling Coordination Degree

To identify the main constraints on FSI coordination, we applied an obstacle diagnosis model, which quantifies the extent to which each factor hinders system development (Wang et al. 2020; Zhao et al. 2024).

$$P_{ij} = 1 - Y_{ij} \quad (6)$$

$$O_{ij} = \frac{P_{ij} * W_i}{\sum_{i=1}^m P_{ij} * W_j} * 100\% \quad (7)$$

where O_{ij} denoted the degree of impairment of the i th indicator in year j , and P_{ij} was the degree of deviation of indicator I in year j . The following table shows the degree of deviation of the indicator I in year j .

2.4.4 | Methodology for Analyzing Forest Sustainability Hotspots

Getis-Ord G_i^* was applied to detect spatial clustering patterns and identify statistically significant FSI hotspot and coldspot regions globally (Rossi and Becker 2019; Lv et al. 2022; Bera et al. 2021).

$$G_i^* = \frac{\sum_{j=1}^n w_{ij}x_{ij} - \bar{x} \sum_{j=1}^n w_{ij}}{S \sqrt{\frac{\sum_{j=1}^n w_{ij}^2 - (\sum_{j=1}^n w_{ij})^2}{n-1}}} \quad (8)$$

$$\bar{x} = \frac{\sum_{j=1}^n x_j}{n} \quad (9)$$

$$S = \sqrt{\frac{(\sum_{j=1}^n x_j)^2}{n} - (\bar{x})^2} \quad (10)$$

In this expression, X_j denoted the attribute value of factor j ; w_{ij} represents the spatial weight between factors i and j ; and G_i^* was the standardized statistic (Z-score). Higher Z-scores indicate tighter clustering of high values (hot spots), whereas lower Z-scores indicate tighter clustering of low values (cold spots). In this case, “high values” refer to high FSI scores, indicating clusters of countries with strong forest sustainability performance, while “low values” represent clusters of low FSI scores, reflecting poor sustainability conditions.

2.4.5 | Correlation Analysis Method Between FSI Indicators and Drivers

The Mantel test was a nonparametric method for assessing the correlation between two matrices and was widely used to examine the relationship between a community distance matrix and environmental variables (Liu et al. 2024; Sunagawa et al. 2015; Renzi et al. 2020). In this study, we applied the Mantel test to evaluate the association between the FSI indicator system and its drivers, using Spearman's rank correlation coefficient (ρ) as the test statistic. The coefficient was calculated as follows:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad (11)$$

where ρ denoted Spearman's correlation coefficient, d_i denoted the difference between the rank values of the i th data pair, and n denoted the total number of observed samples.

2.4.6 | Methodology for Analyzing the Drivers of Forest Sustainability

In this study, a multiple linear regression method based on least squares was used to analyze the drivers affecting the FSI in the years 2010, 2015 and 2020 globally (Sharma et al. 2023). In this study, a total of eight drivers that may affect the FSI were selected to derive a multiple linear regression model:

$$FS = \beta_0 + \beta_1 A_1 + \beta_2 A_2 + \beta_3 A_3 + \beta_4 A_4 + \beta_5 A_5 + \beta_6 A_6 + \beta_7 A_7 + \beta_8 A_8 \quad (12)$$

where $A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8$ are the drivers (Table 2), $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6, \beta_7, \beta_8$ denoted the regression coefficient and ϵ denoted the error variable.

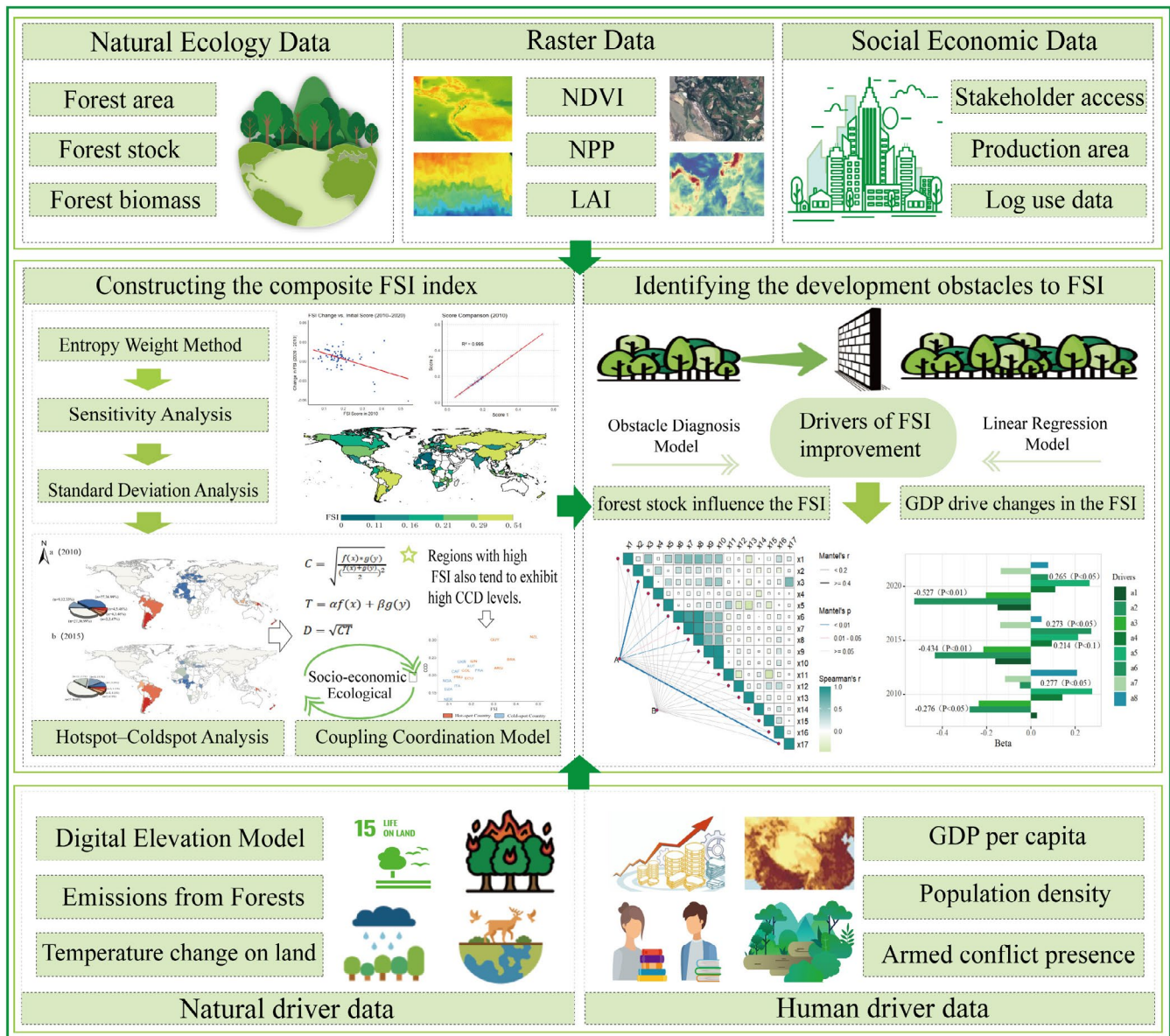


FIGURE 2 | Conceptual framework diagram. (The research framework consists of three main components. First, an integrated FSI was developed by combining natural ecological data, remote sensing data, and socio-economic indicators. A sensitivity analysis was then conducted to ensure the robustness of the index. Second, the spatial distribution of FSI hotspots and coldspots was identified, followed by the calculation of the CCD between ecological and socio-economic sub-systems. Third, an obstacle diagnosis model was applied to detect the key constraints limiting FSI development, and a linear regression model was used to identify the main driving factors influencing FSI at the global scale.)

3 | Results and Analyses

3.1 | Spatial and Temporal Trends in FSI for 73 Countries Globally

Based on the comprehensive evaluation results of FSI for each country, the average FSI across the 73 assessed countries exhibited an overall upward trend, increasing from 0.187 in 2010 to 0.182 in 2015 and 0.191 in 2020. Although some temporal fluctuations were observed, the general trajectory was positive, with 79.6% of countries showing an increase in FSI in 2020 compared to 2010 (Figure 3). The US demonstrated the most substantial improvement, with an FSI increase of 0.06, followed by China and Gabon, each with a gain of 0.03. In contrast, New Zealand

recorded the largest decline, with a decrease of 0.06, and the Philippines followed closely with a reduction of 0.05. Mauritania consistently recorded the lowest FSI during the decade, yet its score showed a notable improvement from 0.04 in 2010 to 0.06 in 2020. Despite a decline in FSI, New Zealand maintained the highest score throughout the study period and remained within the “good” forest sustainability category. Temporally, several forest-rich countries such as China, the United States, Russia, and Argentina advanced from “poor” to “good” FSI categories by 2020, reflecting stronger policy efforts and ecological improvements. Statistical analysis using standard deviation and the Wilcoxon signed-rank test confirmed that these improvements were significant ($p < 0.001$), particularly among countries with lower baseline scores (Figure A2 and Table A3).

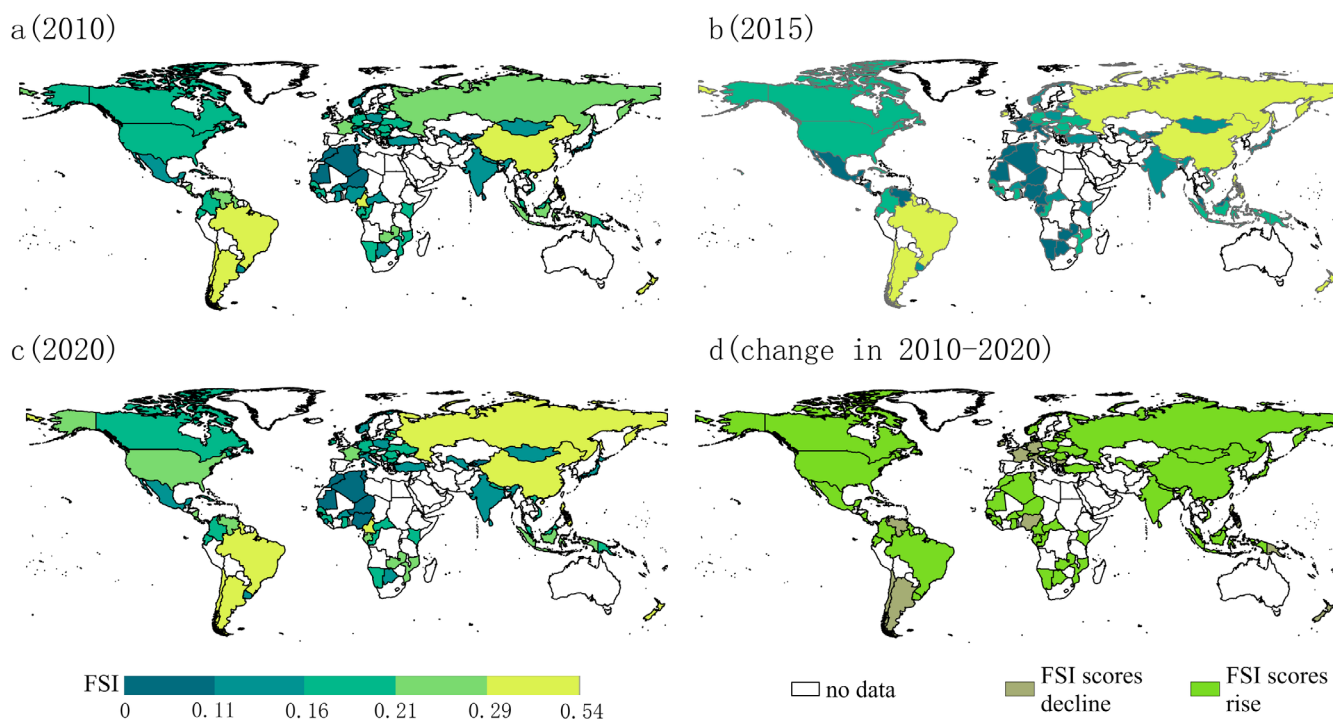


FIGURE 3 | Panels a–c presents the composite scores of the global FSI for 73 countries in 2010, 2015, and 2020, respectively. Panel d illustrates the change in FSI between 2010 and 2020. To ensure consistency in classification across the three time points, we applied the mean values derived from the natural break points method as a uniform grading standard. Based on this method, FSI scores were categorized into five classes: very poor forest sustainability ($0 \leq \text{FSI} \leq 0.11$), poor forest sustainability ($0.11 < \text{FSI} \leq 0.16$), fair forest sustainability ($0.16 < \text{FSI} \leq 0.21$), good forest sustainability ($0.21 < \text{FSI} \leq 0.29$), and very good forest sustainability ($0.29 < \text{FSI} \leq 0.54$).

3.2 | Distribution of FSI Hotspot and Coldspots

This study examines the spatial dynamics of hotspots and coldspots in the FSI across 73 countries for the 2010, 2015, and 2020. Countries were classified using Getis-Ord G_i^* Z-scores, with thresholds of ± 1.65 , ± 1.96 , and ± 2.58 corresponding to 90%, 95%, and 99% confidence levels, respectively. Overall, FSI hotspots were primarily concentrated in South America and Southeast Asia, while coldspots were predominantly located in Northern Africa and Eastern Europe. Key regions with significant changes in FSI levels over time were identified through comparative analysis of the three time points (Figure 4). In 2010, FSI hotspots were mainly situated in the Southern Hemisphere, particularly in South America, where countries such as Argentina and Chile exhibited significantly high FSI at the 99% confidence level. Portions of Southeast Asia also emerged as hotspots, with confidence levels ranging from 90% to 95%.

In contrast, coldspots were primarily observed in Northern Africa, such as Algeria, Libya, and Egypt. By 2015, Argentina remained an FSI hotspot, although now at the 90% confidence level. A notable reduction in coldspot regions was observed in Europe, particularly in Southern and Eastern Europe, where the coldspot significance decreased. However, coldspot countries in Northern Africa persisted, maintaining low FSI. In 2020, FSI hotspot regions in South America became more spatially concentrated and continued to demonstrate high FSI. Additionally, new hotspots emerged in Southeast Asia, notably in Indonesia and Papua New Guinea, indicating substantial improvements in FSI. New coldspot areas appeared in

parts of Northern and Eastern Europe, and the coldspot areas in Northern Africa further expanded, reflecting a continued decline in FSI levels. In contrast, several Eastern European countries, such as Romania and Slovakia, were no longer classified as coldspots in 2015, suggesting temporary improvements. By 2020, African countries including Mauritania and Kenya had fully moved out of coldspot status, indicating notable progress in forest sustainability from previously low baselines.

3.3 | Coupling Coordination and Obstacle Degree in FSI Hotspot and Coldspot Regions

In this study, we applied a CCDM to assess the interaction between the two subsystems underlying the FSI within coldspot and hotspot regions. Due to the limited number of countries identified as coldspots, we selected eight representative countries from each category and compared their average CCD and obstacle degree values across the three study years.

Numerically, the CCD in hotspot regions was significantly higher than those in coldspot regions (Figure 5a). The average CCD in hotspot areas was 0.31, corresponding to *slightly unbalanced development* (Grade 4), while coldspot regions averaged 0.24, indicating *moderately unbalanced development* (Grade 3). Specifically, 63% of coldspot countries fell into the *moderately unbalanced* category ($\text{CCD} = 0.20\text{--}0.29$), whereas 50% of hotspot countries were within the *slightly unbalanced* range ($\text{CCD} = 0.30\text{--}0.39$). Among them, New

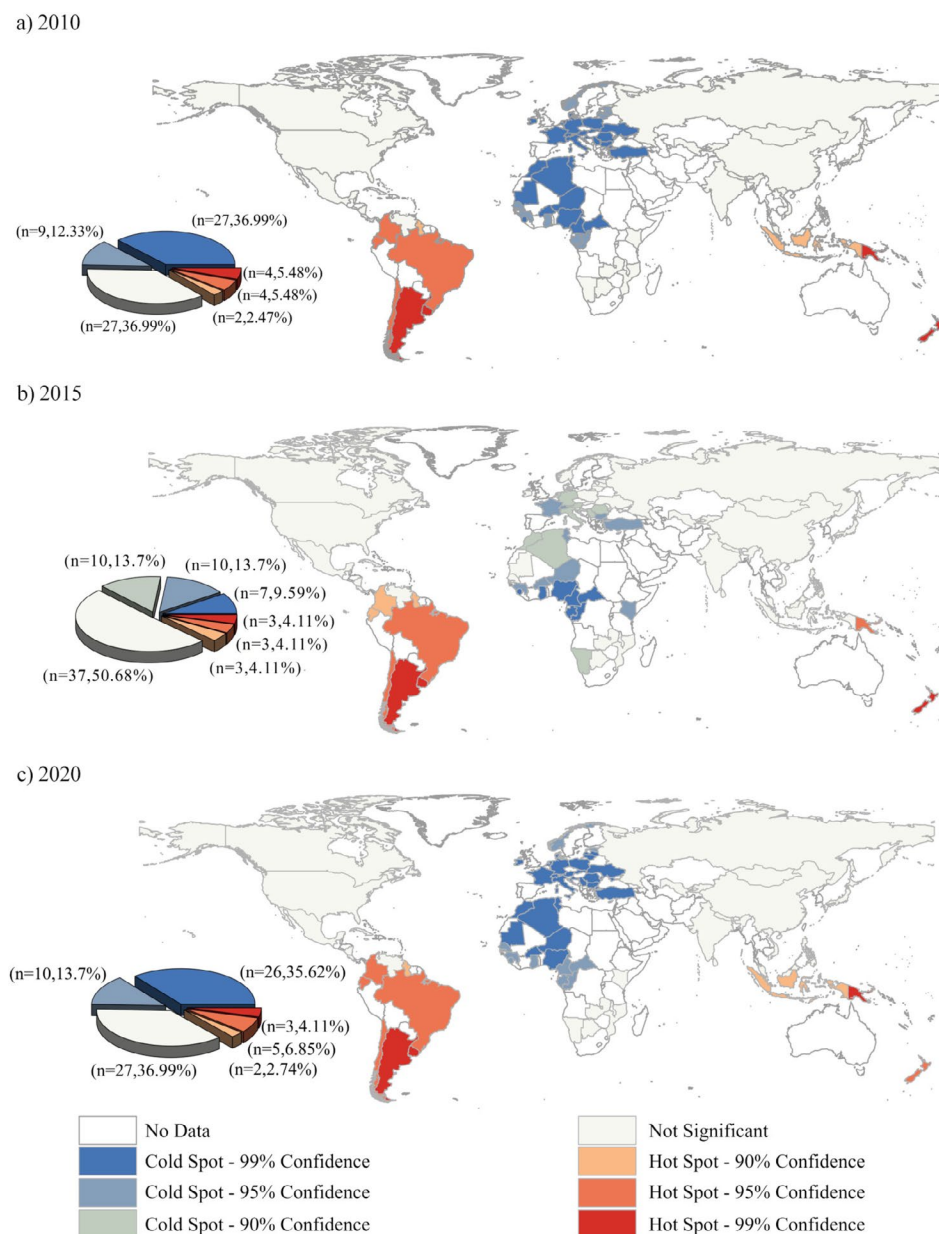


FIGURE 4 | Panels a–c show the global distribution of cold and hot spots in the FSI for 2010, 2015, and 2020, respectively. The accompanying pie charts illustrate the proportion of countries falling within each of the seven statistical confidence intervals, with *n* indicating the number of countries represented in each interval. The color scheme of the pie charts corresponds directly to the legend used for the hotspot map confidence intervals.

Zealand—identified as a hotspot—recorded the highest CCD value in 2010 ($CCD = 0.51$), which falls into *barely balanced development* (Grade 6), approaching the threshold of coordinated development.

These findings confirm the positive link between FSI and CCD, and show that hotspot countries generally maintain a better balance between ecological and socio-economic subsystems. In contrast, coldspot countries often show a clear mismatch between the two. We found that improvement in one subsystem does not necessarily lead to progress in the other. For example, ecological decline is often not compensated by social development, and the reverse is also true. This imbalance may be a key factor limiting overall forest sustainability in these regions.

To further identify the key limiting factors affecting subsystem coordination, we employed an obstacle degree diagnostic model to evaluate the FSI indicator system and isolate the primary obstacle indicators (Figure 5c). Although hotspot and coldspot countries differ in FSI and CCD levels, they do not necessarily face different types of limiting factors (Tables A4 and A5). Across 48 country-level cases selected for obstacle analysis, the most frequently occurring obstacles included:

- x2 (percentage of forested area within protected areas): 42 occurrences (~88%), average obstacle degree: 4.3%
- x3 (forest stock): 45 occurrences (about 94%), average obstacle degree: 25.4%

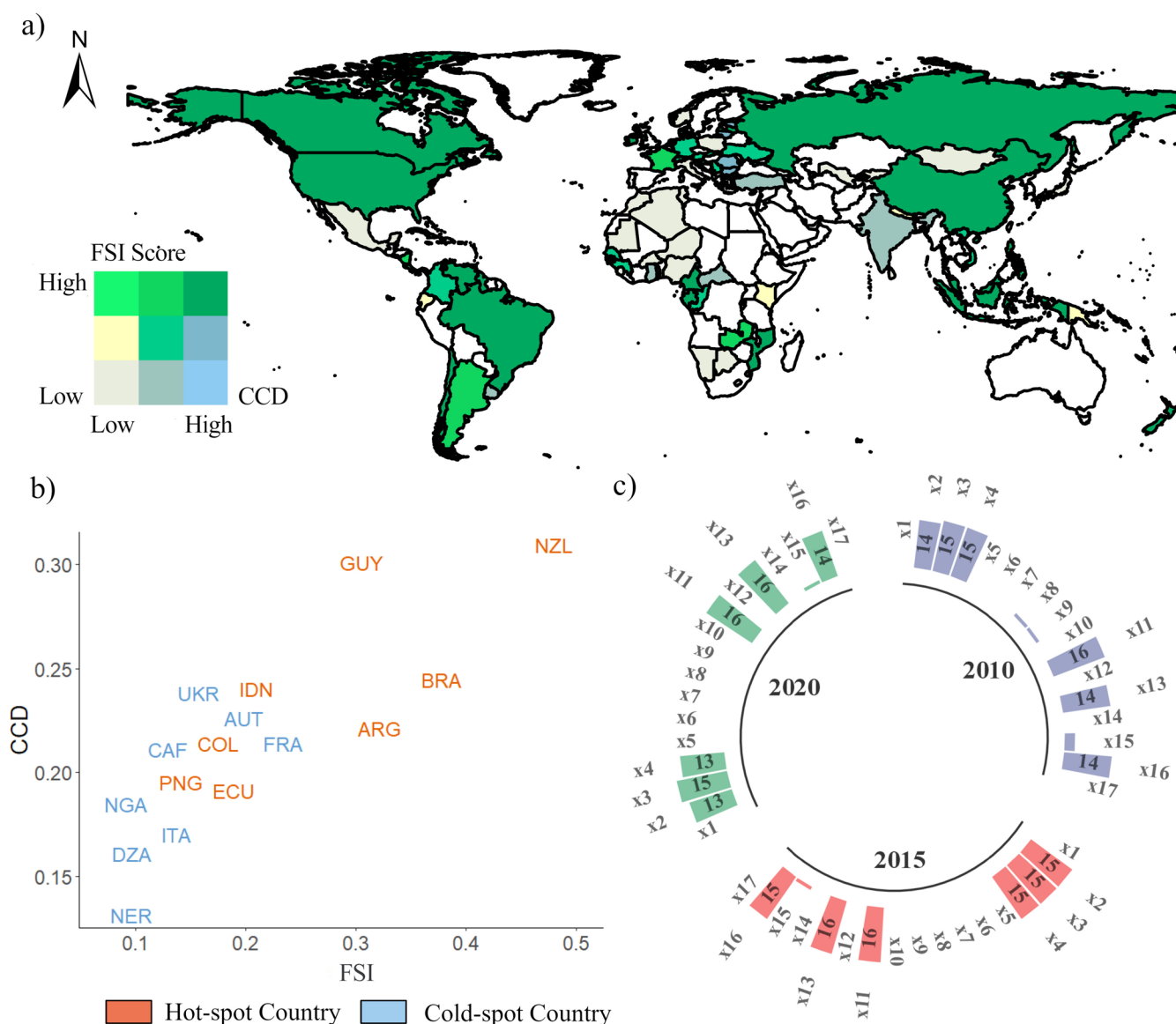


FIGURE 5 | (a) Bivariate plot of FSI versus coupled coordination (where darker green colour indicate larger values of both). (b) Bivariate scatter plot of FSI versus CCD for the 16 coldspot and hotspot countries (hot-spot countries (orange) include Argentina (ARG), Brazil (BRA), Colombia (COL), Ecuador (ECU), Guyana (GUY), Indonesia (IDN), New Zealand (NZL), and Papua New Guinea (PNG). Cold-spot (blue) countries include Austria (AUT), Central African Republic (CAF), Algeria (DZA), France (FRA), Italy (ITA), Nigeria (NGA), Niger (NER), and Ukraine (UKR)). (c) Obstacle diagnostics for 2010, 2015, and 2020, with the highest level of obstruction of x2 (indicating that the protected area is within the forest area as a percentage of forest area), x3 (denoted forest stock), x4 (denoted percentage of forest area targeted for biodiversity conservation), x11 (denoted area of other forested land as a percentage of national land area), x13 (denoted total stock of introduced tree species), and x16 (denoted percentage of forest area targeted for forest production).

- x4 (percentage of forest area designated for biodiversity conservation): 43 occurrences (about 90%), average obstacle degree: 6.5%
- x11 (area of other forested land as a percentage of national territory): 48 occurrences (100%), average obstacle degree: 15.4%
- x13 (total stock of introduced species): 46 occurrences (about 96%), average obstacle degree: 29.9%
- x16 (percentage of forest area designated for forest production): 43 occurrences (about 90%), average obstacle degree: 6.0%.

Notably, introduced species (x13) and low forest stock (x3) consistently ranked as the most severe obstacles, exhibiting the highest average obstacle degrees. These findings support the conclusion that biological pressures and limited stocking capacity are the most critical constraints on forest sustainability coordination. Their high frequency and intensity highlight persistent structural weaknesses in ecological subsystems across countries.

3.4 | Global Driver Factors of Forest Sustainability

This study investigated the relationships between FSI indicators and potential driving factors using Mantel's test and Spearman

correlation analysis. Results revealed statistically significant correlations between natural drivers and FSI indicators in 2010, 2015, and 2020 (Figure 6a–c), with stronger correlations observed in 2010 and 2020 (Mantel's $r > 0.4$). Across all 3 years, natural drivers were significantly associated with forest stock and log consumption (Mantel's $p < 0.01$). Specifically, in 2010, natural drivers also showed significant correlations with the total number of introduced tree species ($p < 0.01$); in 2015, they were more strongly associated with the LAI ($p < 0.01$); and in 2020, they were significantly correlated with the NDVI ($0.01 < p < 0.05$). In contrast, artificial drivers did not exhibit strong correlations with FSI indicators. However, FSI indicators themselves displayed varying degrees of positive intercorrelation. These findings suggest that natural drivers play a more prominent role in influencing changes in forest sustainability.

To further identify key drivers of FSI, multiple linear regression analysis was conducted. The results identified three natural drivers that passed the significance threshold ($p < 0.05$), with A2 (forest-related carbon emissions) and A5 (mean annual precipitation) consistently emerging as the dominant factors throughout

the study period (Figure 6d). Notably, A2 had a significant negative effect on FSI, with an average effect size of -0.41 ($p < 0.01$), increasing from -0.27 ($p < 0.05$) in 2010 to -0.52 ($p < 0.01$) in 2020 (Table A6). This suggests that rising forest emissions significantly constrain forest sustainability. In contrast, A5 exhibited a stable and positive influence on FSI, with an average effect size of 0.25 ($p < 0.01$), indicating that higher levels of precipitation promote improved forest sustainability. Additionally, A6 (GDP per capita) was found to have a significant positive effect on FSI in 2015, with a coefficient of 0.27 ($p < 0.01$), suggesting that greater economic resources may also enhance national forest sustainability.

4 | Discussion

4.1 | Global Trends in Forest Sustainability

The observed upward trend in the FSI aligns with the positive trajectory of global forest development reported in the FAO global forest resources assessment. This overall improvement

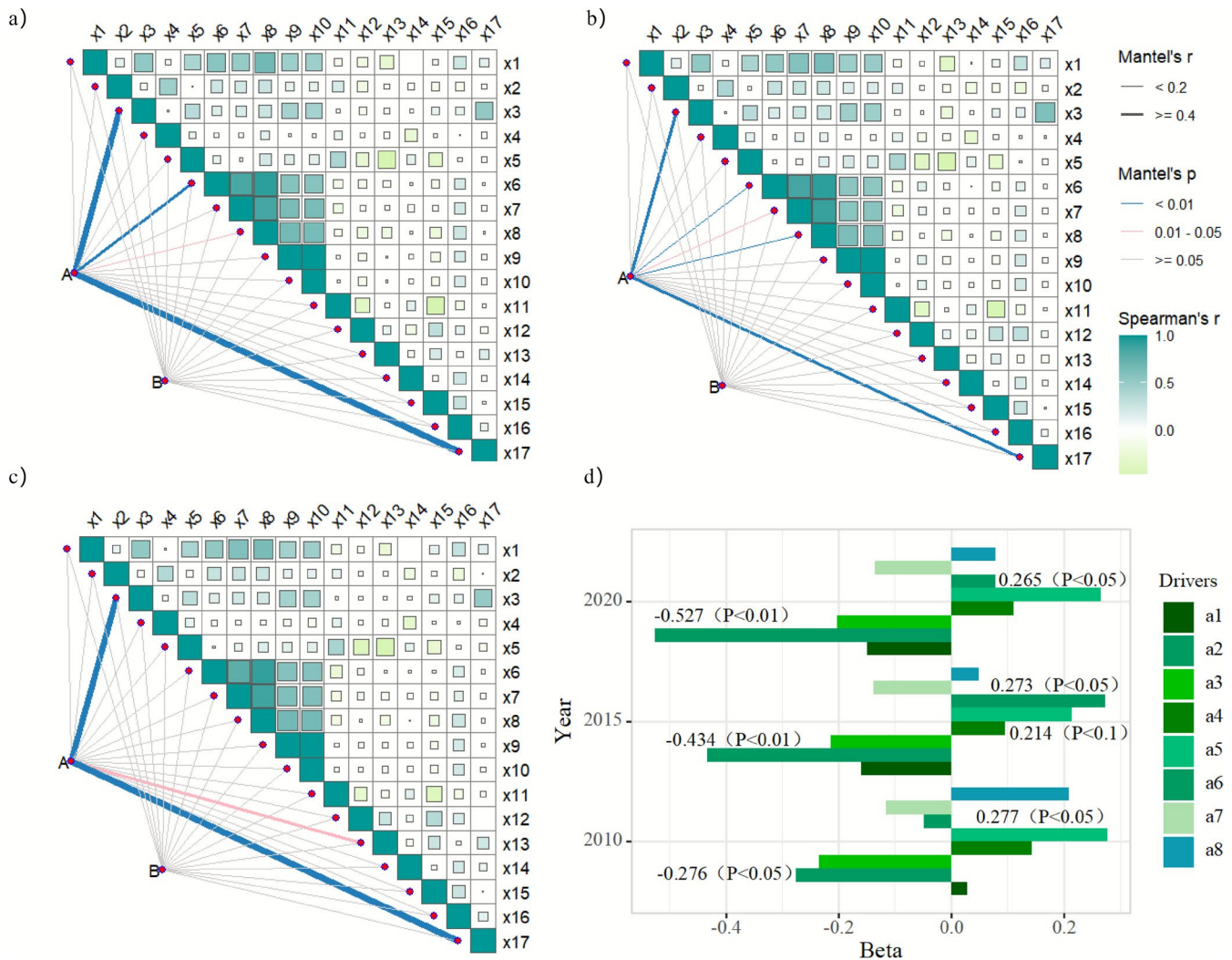


FIGURE 6 | (a–c) Mantel test results for 2010, 2015, and 2020, respectively. A represents natural drivers, and B represents anthropogenic (human-induced) drivers. Shades of green indicate Spearman correlation strength, with darker colors representing stronger correlations. (d) indicating the magnitude and direction of the influence of each driver on the FSI.

reflects the substantial progress in forest protection and sustainability policies. Notably, countries such as China, Russia, and Argentina demonstrated significant increases in FSI by 2020, indicating tangible enhancements in forest quality driven by more proactive conservation and sustainability initiatives. These improvements are closely tied to efforts such as forest protection programs, afforestation, and the promotion of forest carbon sinks. For example, China's large-scale ecological restoration projects, such as the "Grain for Green" program and national afforestation campaigns, have likely played a crucial role in the observed FSI gains (Huang et al. 2020).

Globally, FSI exhibits pronounced spatial heterogeneity at both continental and national scales. Among long-standing FSI hotspot countries, Argentina and Chile stand out due to effective forest governance and legal frameworks. For instance, Argentina's enactment of the 2007 Forestry Law has strictly regulated forest exploitation and promoted sustainable management (Lende 2018). In Southeast Asia, emerging hotspot countries such as Indonesia have made progress by curbing illegal logging through the V-LEGAL certification system, advancing the Indonesian forest and climate action plan (IFMAP) to restore degraded forests, and enhancing regional cooperation to combat cross-border forest crimes.

Conversely, FSI coldspot countries continue to face structural and policy-related obstacles. Eastern Europe faces challenges from economic transition and forest overuse, hindering long-term sustainability (Winkel et al. 2022). Similarly, North African nations such as Libya and Egypt contend with extremely limited forest resources, low forest quality, and small forested areas. In Middle East and North Africa countries, political instability, conflict, and poverty have further weakened forest governance capacities (Guo et al. 2019; Mitchard and Flintrop 2013). Notably, New Zealand and Mauritania are outliers—New Zealand maintained the highest FSI despite a slight decline, while Mauritania showed steady gains from a low baseline. These contrasting trends, likely shaped by region-specific drivers, warrant further investigation.

To enhance the interpretive robustness of the FSI, it is also important to consider its alignment with international sustainable forest management frameworks. The FSI reflects similar priorities to initiatives such as the Montreal Process and Forest Europe, which emphasize the integration of ecological, economic, and social dimensions. The FSI is also relevant for monitoring SDG 15.2 on sustainable forest management. Unlike conventional SDG indicators that primarily rely on forest area and biomass stock, the FSI offers a more nuanced and multidimensional perspective by incorporating both ecological health and socio-economic coordination. Despite data gaps and methodological limitations, the FSI may complement existing SDG monitoring efforts by providing deeper insights into national forest sustainability performance and cross-country comparability.

4.2 | Distribution of Hotspot and Coldspot in Forest Sustainability

Our analysis revealed stronger coordination between subsystems in FSI hotspot regions, which also exhibited significantly higher

forest sustainability. Through obstacle factor diagnostics, we identified key limiting indicators, including the total number of introduced tree species, forest stock, and the proportion of other forestry. These factors consistently constrained subsystem coordination during 2010–2020. Forest stock is a critical indicator of forest health (Shah et al. 2024). Its persistently low levels in many coldspot regions highlight inefficiencies in forest resource use and the need for improved forest management. The proportion of other forested land serves as a proxy for forest spatial distribution. In many developing countries in Africa and Asia, forest conditions can be improved through targeted interventions such as afforestation, ecological restoration projects, and land conversion initiatives (e.g., returning farmland to forest).

We identified external driving factors influencing the FSI beyond the indicator system itself. Among the anthropogenic drivers selected for analysis, only per capita GDP exhibited a statistically significant impact on the FSI. These findings suggest that while human activities undoubtedly influence forest sustainability, their effects are modulated by complex context-dependent interactions, especially at a global scale. Relying solely on anthropogenic factors may therefore be insufficient to fully capture the dynamics of forest sustainability. These findings reinforce the central role of natural drivers in shaping global FSI trends (Crouzeilles et al. 2016), offering valuable insights for future forest policy and management strategies.

4.3 | Differences in Forest Sustainability Drivers

Among the drivers examined, the negative correlation between A2 (forest emission levels) and FSI is particularly notable (MacBean et al. 2022). Elevated forest emissions significantly constrain forest sustainability by degrading ecological structures and compromising forest health and stability. These findings underscore the urgency of implementing stringent forest emission control policies aimed at reducing harmful outputs, preserving ecological integrity, and promoting long-term forest resilience (Ge and Lin 2024).

In contrast, A5 (average annual precipitation) showed a consistently positive correlation with FSI, reflecting its essential role in supporting forest growth, productivity, and biodiversity (Bonan 2008; Bruijnzeel 2004; Pérez-Ruiz et al. 2018). These results are consistent with previous studies (e.g., Crouzeilles et al. 2016), which emphasize the dominant role of environmental context in shaping sustainability outcomes. However, increasing rainfall variability under climate change adds uncertainty, making stable hydrological systems crucial for long-term forest sustainability (Chen et al. 2025).

Among human-related factors, A6 (GDP per capita) exhibited a positive association with FSI. High-income countries like the USA and France have gained forest cover, supported by sound policy and financial capacity. However, many tropical countries continue to face deforestation pressures, revealing the dual role of economic development as both a risk and opportunity depending on governance and land-use dynamics (Hoang and Kanemoto 2021). This duality limits the explanatory power of GDP when analyzed in isolation.

The limited explanatory power of demographic factors, such as population density, may stem from their indirect and context-dependent effects. While high population pressure can drive deforestation, it may also lead to institutional innovation under favorable governance. These findings reinforce that regression results reflect correlations, not causation, and should be interpreted within broader structural contexts. Ultimately, natural drivers like precipitation and emissions show stronger global associations with FSI, but effective governance remains crucial to translating ecological potential into sustainable outcomes, particularly in socially or politically constrained regions.

4.4 | Limitations and Outlook

A key limitation of this study lies in the availability of long-term, consistently collected forest sustainability data. Due to issues of data completeness and consistency, we selected representative datasets from 73 countries with relatively reliable forest statistics. While this enables cross-country comparison, it excludes some non-forested or data-deficient countries and limits year-by-year trend analysis. The five-year interval of FAO's Global Forest Resources Assessments constrains the capture of finer temporal changes. Although the United Nations Commission on Sustainable Development (UNCSD) has proposed indicators such as logging intensity and forest area change, these face challenges like inconsistent time series, limited accessibility, and difficulty in accurately distinguishing natural forest cover (Nabhani et al. 2024). Moreover, short-term disturbances such as wildfires or policy shocks may not be fully reflected due to the temporal resolution of the dataset. Future research should prioritize the development of a globally unified FSI, supported by long-term, high-quality data across countries, with more frequent updates and higher spatial granularity.

Another limitation concerns the scope of indicators. Although we adopted a multi-dimensional approach combining ecological and socio-economic metrics, some important institutional factors were not included in the quantitative framework. For example, while stakeholder participation (x14) reflects certain governance aspects, indicators such as enforcement capacity, corruption levels, or tenure security were not incorporated due to lack of consistent global data. These governance-related factors are known to affect forest outcomes but remain difficult to quantify across countries (Winkel et al. 2022; Hasnaoui and Krott 2019). Furthermore, the assumption of equal importance between the ecological and socio-economic subsystems in the CCDM is a necessary simplification for comparative analysis. However, this approach may overlook contextual differences, such as situations where ecological constraints play a more dominant role than socio-economic factors or the reverse. Future research should seek to integrate more robust governance and institutional indicators, potentially through mixed-methods approaches or regionally tailored frameworks to provide a more comprehensive understanding of the socio-political drivers influencing forest sustainability.

To improve future assessments, there is a need to establish more standardized, long-term forest datasets that include both

biophysical and governance indicators. Combining satellite-based remote sensing with regional field surveys can help fill data gaps in underreported areas. In addition, integrating ecosystem models (e.g., PnET-II) with land use models (e.g., ABM/LUCC) would support scenario analysis of forest change (Dai et al. 2016; Wu et al. 2017). Aligning indicator systems with SDG 15.2 targets could also enhance policy relevance and international comparability. It is important to note that the regression and Mantel test results provide insight into spatial associations between forest sustainability and its potential drivers, but they should not be interpreted as evidence of direct causation. The relationships among ecological, economic, and demographic factors are often complex and mutually influenced, requiring careful interpretation within a broader contextual framework.

5 | Conclusion

The global decline in forest resources has triggered a cascade of environmental crises, including increased forest fragmentation, reduced carbon sequestration, and the degradation of essential ecosystem services. These challenges underscore the urgent need for more systematic and sustainable forest management practices to confront the accelerating impacts of climate change. In response, this study proposes a novel FSI integrating both natural ecological and socio-economic dimensions, enabling the first quantitative, cross-national assessment of forest sustainability across 73 countries from 2010 to 2020.

Although FSI levels varied across countries, the global trend from 2010 to 2020 indicated steady improvement, suggesting enhanced forest ecosystem integrity and better alignment with socio-economic development. Spatial patterns showed that countries such as Argentina and Chile consistently maintained high FSI scores, likely benefiting from stable forest policies. In contrast, countries in Northern Africa and Eastern Europe remained in low-FSI clusters, often due to political instability and limited investment in forest resources. The coordination between ecological and socio-economic subsystems was generally stronger in regions with higher FSI. Factors such as forest emissions, log consumption, and low forest stock emerged as key constraints in low-performing regions. Among all drivers examined, natural factors, particularly precipitation and forest stock, had the most consistent influence on forest sustainability, while human-related factors, such as GDP per capita, showed weaker and more variable effects.

As the 2030 target for the SDGs approaches, we provide a scientific foundation for developing tailored forest conservation strategies. Strengthening adaptive, region-specific management approaches aligned with both ecological conditions and socio-economic capacities will be essential for securing the long-term health of global forests and achieving SDG 15.2. These findings also contribute to evaluating the SDGs forest-related SDG sub-indicators, such as SDG 13 (Climate Change) and SDG 15 (Life on land), to make more governments and organizations pay attention to forest ecosystem and socio-economic system sustainable development.

Author Contributions

Zian Liu: conceptualization, methodology, formal analysis, visualization, software, writing—original draft. **Chengyuan Wang:** formal analysis, software, methodology, writing—review and editing. **Zhenglin Tian:** formal analysis, software, methodology. **Tongpeng Wang:** formal analysis, software, methodology. **Longxin Ding:** writing—review and editing. **Hongpeng Liu:** conceptualization, methodology, writing—review and editing. **Wenyi Fan:** writing—review and editing. **Mingze Li:** writing—review and editing. **Xuegang Mao:** conceptualization, writing—review and editing, supervision, funding acquisition, project administration, resources.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The forest-related data used in this study were obtained from the Food and Agriculture Organization (FAO) Forest Resources Assessment (<https://www.fao.org/forest-resources-assessment/en/>). Normalized difference vegetation index (NDVI) data were sourced from the MOD13A2.061 product via NASA's Land Processes Distributed Active Archive Center (LP DAAC) (<https://lpdaac.usgs.gov/products/mod13a2v061/>). Leaf area index (LAI) data were retrieved from the MOD15A2H.061 product, also via LP DAAC (<https://lpdaac.usgs.gov/products/mod15a2hv061/>). Net primary productivity (NPP) data were accessed through NASA's MOD17A2H.061 product via LP DAAC (<https://lpdaac.usgs.gov/products/mod17a2hv061/>). Elevation data (DEM) were obtained from the Copernicus Global Digital Elevation Model (GLO-30) available through the Copernicus Open Access Hub (<https://spacedata.copernicus.eu/web/cscda/dataset-details?articleId=394198>). Additional variables—such as emissions from forests, temperature change on land, forest loss due to fire, and log consumption—were collected from the FAO FAOSTAT database (<https://www.fao.org/faostat/en/>). Average annual precipitation data were obtained from Google Earth Engine, specifically using the CHIRPS precipitation dataset (https://developers.google.com/earth-engine/datasets/catalog/UCSB_CHG_CHIRPS_DAILY). Socio-economic indicators including GDP per capita and population density were sourced from the World Bank Open Data portal (<https://data.worldbank.org/>). Information on the occurrence of armed conflict was retrieved from Our World in Data—Conflict dataset (<https://ourworldindata.org/war-and-peace>).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.